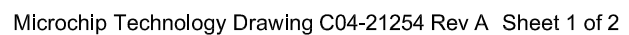
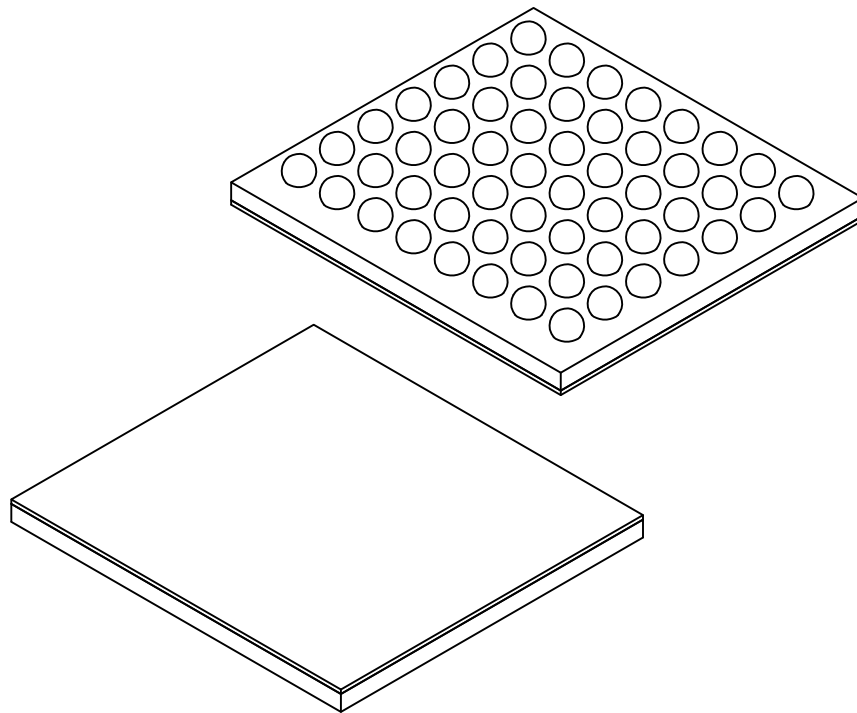


Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



56-ball Wafer Level Chip Scale Package (FZB) - 3.170x3.444x0.483 mm Body [WLCSP] Atmel Legacy GPC GBL

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



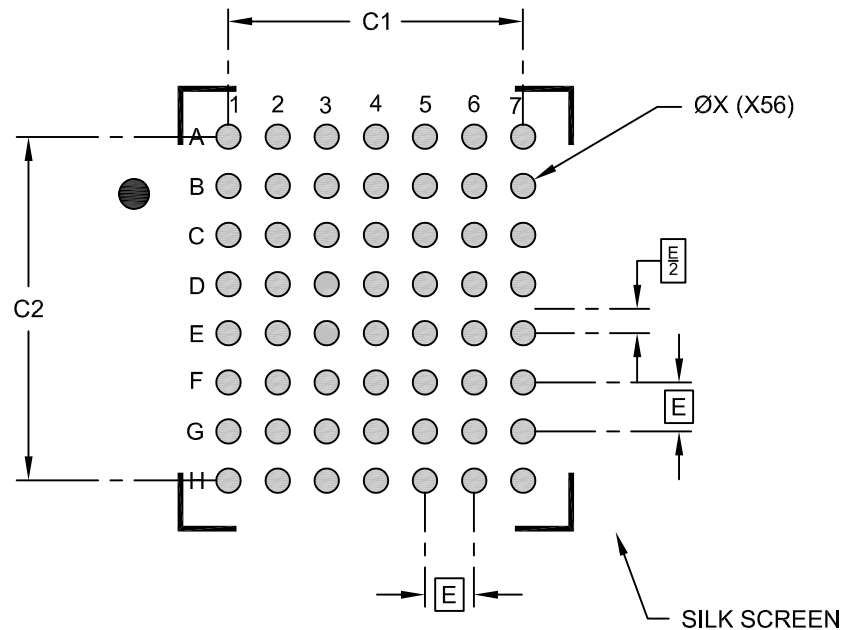
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		56		
Pitch	eD		0.40 BSC		
Pitch	eE		0.40 BSC		
Overall Height	A		0.403	0.443	0.483
Ball Height	A1		0.17	-	0.23
Wafer Thickness	A2		0.178	0.203	0.228
Backside Coat Thickness	A3		0.040 REF		
Overall Length	D		3.170 BSC		
Ball Array Length	D1		2.400 BSC		
Overall Width	E		3.444 BSC		
Ball Array Width	E1		2.800 BSC		
Ball Width	b		0.23	0.26	0.29
Ball Diameter			0.25 REF		

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

56-ball Wafer Level Chip Scale Package (FZB) - 3.170x3.444x0.483 mm Body [WLCSP] Atmel Legacy GPC GBL

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Center Ball Diameter (X56)	X			0.200
Contact Pad Spacing	C1		2.40	
Contact Pad Spacing	C2		2.80	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23254 Rev A